**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 197 · Stacks and queues

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a[1] for int[] a={6,9,3}?

Q2. What is −1 in 8-bit two's complement?

Q3. A loop running from 0 to 3 inclusive executes how many times?

Q4. Push 2, push 5, pop once: which value is removed?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

stack — LIFO abstract data type.

push — Add an item onto a stack.

pop — Remove the top stack item.

queue — FIFO abstract data type.

dequeue — Remove the front queue item.

## New learning

Push — Adding an item puts it on top of a stack.

Pop — Removing an item takes the most recently pushed item: 7.

Contrast with a queue — A queue removes the oldest item instead (FIFO); it is suited to arrival-order processing.

## Worked example

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

1. Predict — The newest value 5 is on top, so Pop removes 5.

2. Confirm and explain — The stack remains [3,8] from bottom to top.

# We Do · Practise with guidance

A queue receives A, B, C, then Dequeue once. Which item leaves and what remains?

```
var queue = new System.Collections.Generic.Queue<char>();
queue.Enqueue('A');
queue.Enqueue('B');
queue.Enqueue('C');
char removed = queue.Dequeue(); // 'A'
```

Predict

The earliest item A is at the front, so it is removed first.

Confirm and explain

The remaining queue is B then C (front to rear).

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Push A, B, C onto a stack then Pop twice. What remains?

### Question 2

Enqueue 10, 20, 30; dequeue; enqueue 40. What remains front-to-rear?

### Question 3

Explain why an undo history usually uses a stack rather than a queue.

# Exit Ticket · Answer before checking

A queue removes the most recently inserted item first.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. 9.

Q2. 11111111.

Q3. 4.

Q4. 5, the last item pushed.

### You Do pages · Independent questions

Q1. A remains. The pops remove C then B.

Q2. 20, 30, 40.

Q3. The most recent change must be undone first, which follows LIFO; FIFO would undo oldest changes first.

### Exit Ticket · Model answer

False: a queue is FIFO, removing the earliest inserted item.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**